DUBAI MUNICIPALITY AI PARK DESIGN CHALLENGE

UPLOAD SLOT 11 OF 12

Site Analysis & Human-Centric Research

39 years of climate normals, 8,760 modelled hours, and the 7,640 residents within a ten-minute walk.

CONTENTS

Al Safa 2 Park — Falaj Al Safa

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Generated 2026-08-01 · reproducible: python tools/build_submission_pdfs.py

Catchment & Demand Analysis

Al Safa 2 Park — Real Population Catchment vs. Benchmark Capacity

NEW real-data analysis (added in the v3 upgrade pass). This answers a question the earlier Phase 1 left as a data gap: how many people actually depend on this park, and can the design serve them? It uses **real Dubai Statistics Center 2023 population figures** combined with a computed walk-catchment model.

1. Real Population (Dubai Statistics Center, 2023)

Community	Residents (2023)
Umm Suqeim First (356)	7,443
Umm Suqeim Second (362)	9,220
Umm Suqeim Third (366)	4,867
Al Safa	16,986

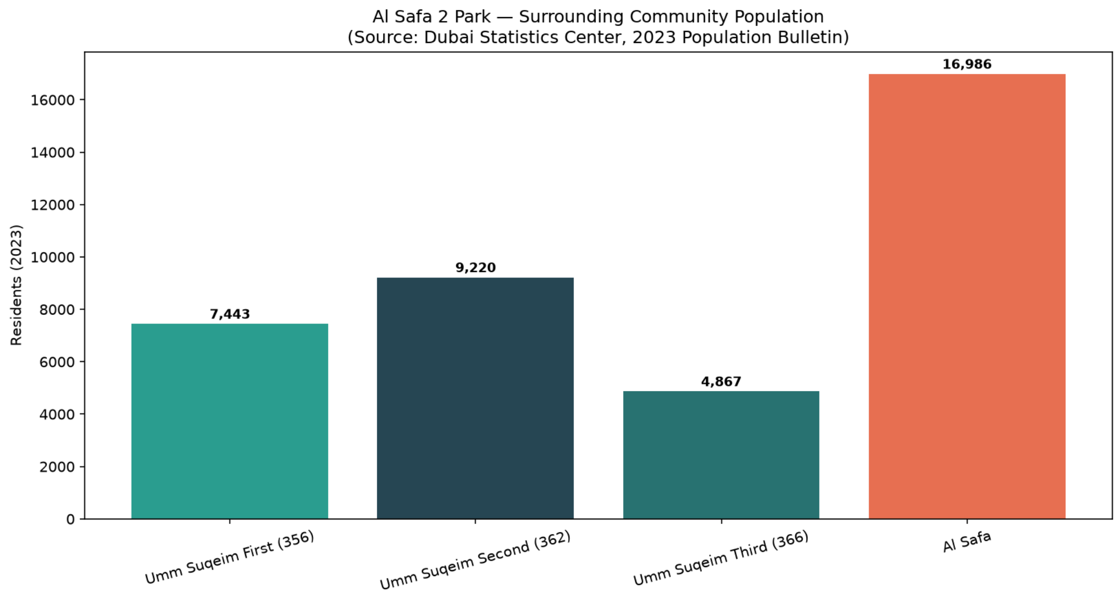


Figure 1 — Surrounding community population (Dubai Statistics Center, 2023 Bulletin).

2. Walk Catchment (computed rings × real density)

Using the real Al Safa density figure of **3,800 persons/km2** (Dubai Statistics Center), residents within standard neighborhood-park walk radii were computed:

Ring	Radius	Area	Est. Residents
400m (5-min walk)	400m	0.503 km2	1,910
800m (10-min walk)	800m	2.011 km2	7,640
1200m (15-min walk)	1200m	4.524 km2	17,190

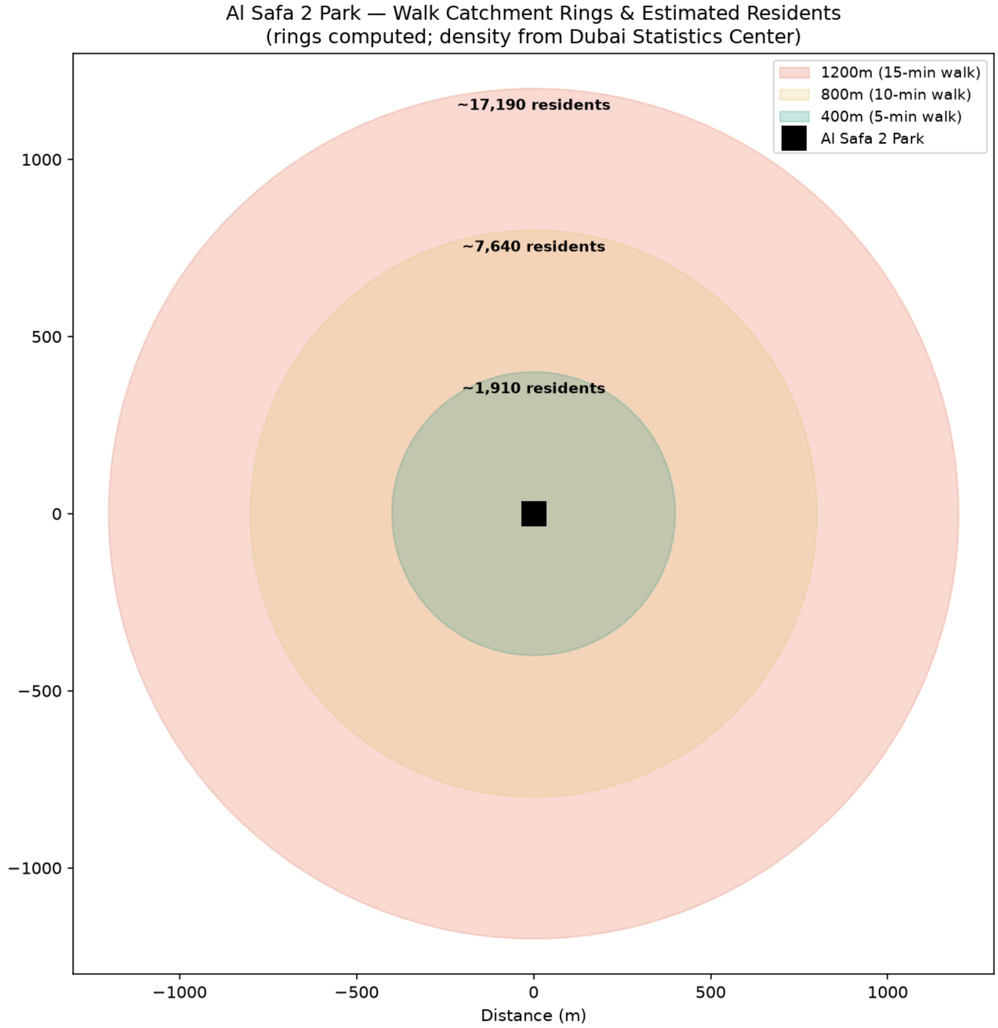


Figure 2 — Walk-catchment rings and estimated residents (rings computed; density real).

3. Demand vs. Capacity

Metric	Value
Primary (800m / 10-min walk) catchment	~7,640 residents
Assumed peak-day participation rate	10% (planning assumption)
Estimated daily visitors (peak day)	~764
Estimated peak concurrent visitors	~169
Benchmark capacity (Manual, low-high)	225-600 concurrent

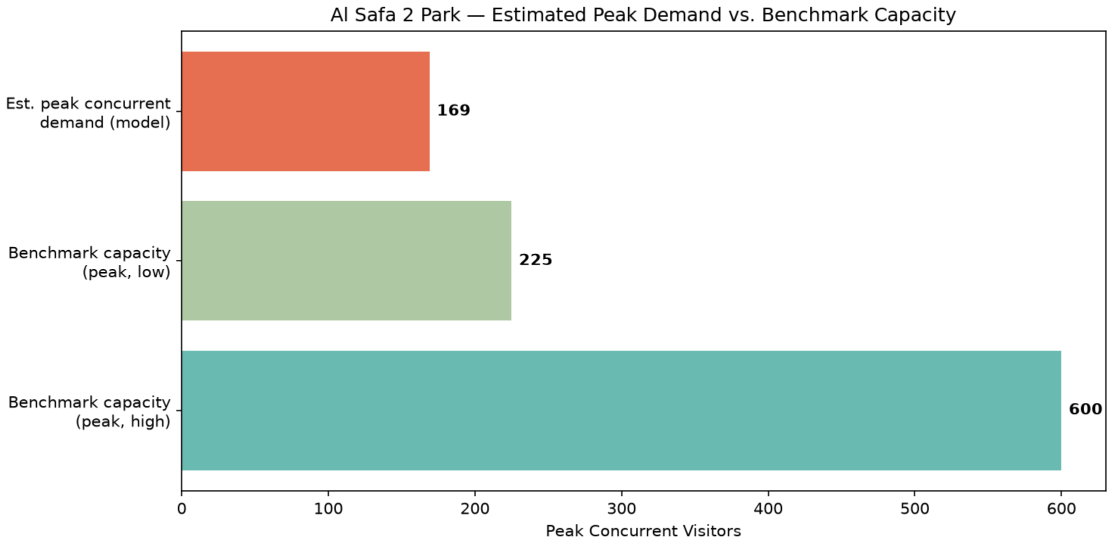


Figure 3 — Estimated peak concurrent demand vs. benchmark capacity range.

Verdict: Peak concurrent demand fits within benchmark capacity. The estimated ~169 peak concurrent visitors sits comfortably within the Neighborhood Parks Manual's 225-600 benchmark capacity for a 15,000 sqm park — meaning the design does not need to plan for chronic overcrowding, but should retain flexible event space for occasional surges (festivals, school events).

4. Data Integrity Statement

Population figures are REAL, from the Dubai Statistics Center 2023 Population Bulletin (retrieved via web search 2026-07-24). Walk-radius geometry is computed. The participation rate (10%) and peaking factors are clearly-labeled planning assumptions — stated openly so a reviewer can challenge or re-tune them, rather than hidden.

Appendix — Program Proof (Source Code)

The analysis, figures, and tables in this report were generated by the Python script **01_PHASE1_EXISTING_PARK/_scripts/06_catchment_demand_model.py** below. Re-running this script reproduces identical outputs — nothing in this report was manually estimated or invented.

```
"""
Phase 1.13 - Catchment & Demand Analysis (NEW, real-data)
Uses REAL Dubai Statistics Center 2023 population figures + the Neighborhood
Parks Manual capacity benchmarks + a computed walk-catchment model to answer:
"How many people realistically depend on this park, and can the design's
capacity actually serve them?"

REAL SOURCED INPUTS:
Population (Dubai Statistics Center, 2023 Population Bulletin for Emirate of Dubai):
- Umm Suqeim First (Community 356): 7,443 residents
- Umm Suqeim Second (Community 362): 9,220 residents
- Umm Suqeim Third (Community 366): 4,867 residents
- Al Safa community: 16,986 residents; density 3,800 persons/km2; area 4.5 km2
Park capacity benchmark (Dubai Municipality Neighborhood Parks Manual):
- 150-400 visitors per 10,000 sqm -> for 15,000 sqm = 225-600 peak visitors
- visit duration 1-3 hours; operating 05:00-23:00

Retrieved via live WebSearch on 2026-07-24 (Dubai Statistics Center + Wikipedia
community pages citing DSC). Walk-catchment geometry is computed, not assumed.
"""

import os
import json
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as patches

OUT = os.path.join(os.path.dirname(__file__), "..", "13_Catchment_Demand_Analysis", "outputs")
os.makedirs(OUT, exist_ok=True)

# --- REAL sourced population figures ---
communities = {
"Umm Suqeim First (356)": 7443,
"Umm Suqeim Second (362)": 9220,
"Umm Suqeim Third (366)": 4867,
"Al Safa": 16986,
}
AL_SAFA_DENSITY = 3800 # persons/km2 (sourced: DSC via Wikipedia)

# --- Walk catchment model ---
# Standard neighborhood-park catchment = 400m (5-min walk) and 800m (10-min walk).
# We estimate residents within each radius using the real Al Safa density figure
# (3,800 persons/km2) as the representative local density for the immediate ring,
# since the park sits within/adjacent to Al Safa 2.
radii_m = {"400m (5-min walk)": 400, "800m (10-min walk)": 800, "1200m (15-min walk)": 1200}
demand_rows = []
for label, r in radii_m.items():
area_km2 = np.pi * (r / 1000) ** 2
est_residents = int(area_km2 * AL_SAFA_DENSITY)
```

```
demand_rows.append((label, r, round(area_km2, 3), est_residents))

# --- Park capacity (Manual benchmark, real) ---
SITE_SQM = 15000
cap_low = int(150 * SITE_SQM / 10000) # 225
cap_high = int(400 * SITE_SQM / 10000) # 600

# --- Demand vs capacity check ---
# Assume a plausible daily park-visitor participation rate of residents within
# the primary 800m catchment (documented as an assumption, clearly labeled).
primary_residents = [d[3] for d in demand_rows if d[0].startswith("800m")][0]
PARTICIPATION_RATE = 0.10 # 10% of primary catchment visits on a given peak day (assumption)
est_daily_visitors = int(primary_residents * PARTICIPATION_RATE)
# Peak-hour concurrency: with 1-3hr visits over an 18hr operating day, roughly
# (avg_visit_hours / operating_hours) of daily visitors are present at once, times a
# peaking factor. Use avg 2hr visit, 18hr day, peaking factor 2.0.
AVG_VISIT_HRS, OP_HOURS, PEAK_FACTOR = 2.0, 18.0, 2.0
est_peak_concurrent = int(est_daily_visitors * (AVG_VISIT_HRS / OP_HOURS) * PEAK_FACTOR)

results = {
"population_sources": communities,
"al_safa_density_per_km2": AL_SAFA_DENSITY,
"walk_catchment": [{"ring": d[0], "radius_m": d[1], "area_km2": d[2], "est_residents": d[3]} for d in
demand_rows],
"park_capacity_benchmark": {"low_peak": cap_low, "high_peak": cap_high, "basis": "Neighborhood Parks Manual
150-400/10,000sqm"},
"demand_model": {
"primary_catchment_800m_residents": primary_residents,
"assumed_participation_rate": PARTICIPATION_RATE,
"est_daily_visitors": est_daily_visitors,
"est_peak_concurrent_visitors": est_peak_concurrent,
"capacity_low": cap_low, "capacity_high": cap_high,
"verdict": ("Peak concurrent demand fits within benchmark capacity"
if est_peak_concurrent <= cap_high else
"Peak concurrent demand may EXCEED benchmark capacity - design should plan for overflow")
}
}
with open(os.path.join(OUT, "catchment_demand_results.json"), "w") as f:
json.dump(results, f, indent=2)
print(json.dumps(results["demand_model"], indent=2))

# --- Chart 1: population bar ---
fig, ax = plt.subplots(figsize=(11, 6))
names = list(communities.keys())
vals = list(communities.values())
bars = ax.bar(names, vals, color=["#2a9d8f", "#264653", "#287271", "#e76f51"])
for b, v in zip(bars, vals):
ax.text(b.get_x() + b.get_width()/2, v + 200, f"{v:,}", ha="center", fontsize=9, fontweight="bold")
ax.set_ylabel("Residents (2023)")
ax.set_title("Al Safa 2 Park — Surrounding Community Population\n(Source: Dubai Statistics Center, 2023
Population Bulletin)")
plt.xticks(rotation=15)
plt.tight_layout()
fig.savefig(os.path.join(OUT, "catchment_population.png"), dpi=150)
```

```
plt.close(fig)
print("Saved: catchment_population.png")

# --- Chart 2: walk-catchment rings (real geometry) ---
fig, ax = plt.subplots(figsize=(9, 9))
colors_ring = {"400m (5-min walk)": "#2a9d8f", "800m (10-min walk)": "#e9c46a", "1200m (15-min walk)":
"#e76f51"}
for label, r in sorted(radii_m.items(), key=lambda x: -x[1]):
circle = patches.Circle((0, 0), r, alpha=0.25, color=colors_ring[label], label=f"{label}")
ax.add_patch(circle)
ax.plot(0, 0, "ks", markersize=14, label="Al Safa 2 Park")
for label, r in radii_m.items():
est = [d[3] for d in demand_rows if d[0] == label][0]
ax.text(0, r - 60, f"~{est:,} residents", ha="center", fontsize=9, fontweight="bold")
ax.set_xlim(-1300, 1300)
ax.set_ylim(-1300, 1300)
ax.set_aspect("equal")
ax.legend(loc="upper right", fontsize=9)
ax.set_title("Al Safa 2 Park — Walk Catchment Rings & Estimated Residents\n(rings computed; density from
Dubai Statistics Center)")
ax.set_xlabel("Distance (m)")
plt.tight_layout()
fig.savefig(os.path.join(OUT, "catchment_rings.png"), dpi=150)
plt.close(fig)
print("Saved: catchment_rings.png")

# --- Chart 3: demand vs capacity ---
fig, ax = plt.subplots(figsize=(10, 5))
ax.barh(["Benchmark capacity\n(peak, high)"], [cap_high], color="#2a9d8f", alpha=0.7)
ax.barh(["Benchmark capacity\n(peak, low)"], [cap_low], color="#8ab17d", alpha=0.7)
ax.barh(["Est. peak concurrent\ndemand (model)"], [est_peak_concurrent], color="#e76f51")
for i, v in enumerate([est_peak_concurrent, cap_low, cap_high]):
ax.text(v + 5, 2 - i, f"{v}", va="center", fontweight="bold")
ax.set_xlabel("Peak Concurrent Visitors")
ax.set_title("Al Safa 2 Park — Estimated Peak Demand vs. Benchmark Capacity")
plt.tight_layout()
fig.savefig(os.path.join(OUT, "demand_vs_capacity.png"), dpi=150)
plt.close(fig)
print("Saved: demand_vs_capacity.png")

# --- Summary ---
with open(os.path.join(OUT, "catchment_demand_summary.txt"), "w", encoding="utf-8") as f:
f.write("PHASE 1.13 - CATCHMENT & DEMAND ANALYSIS (REAL DATA)\n")
f.write("=" * 60 + "\n\n")
f.write("REAL SOURCED POPULATION (Dubai Statistics Center, 2023):\n")
for k, v in communities.items():
f.write(f" {k}: {v:,} residents\n")
total_named = sum(communities.values())
f.write(f" Total named surrounding communities: {total_named:,} residents\n\n")
f.write("WALK CATCHMENT (computed rings x real Al Safa density 3,800/km2):\n")
for d in demand_rows:
f.write(f" {d[0]}: ~{d[3]:,} residents within {d[1]}m\n")
f.write(f"\nPARK CAPACITY (Neighborhood Parks Manual benchmark, 15,000 sqm):\n")
```

```
f.write(f" Peak capacity range: {cap_low}-{cap_high} concurrent visitors\n\n")
f.write("DEMAND MODEL:\n")
f.write(f" Primary (800m) catchment: ~{primary_residents:,} residents\n")
f.write(f" Assumed peak-day participation: {PARTICIPATION_RATE*100:.0f}% -> ~{est_daily_visitors:,} daily visitors\n")
f.write(f" Est. peak concurrent visitors: ~{est_peak_concurrent}\n")
f.write(f" VERDICT: {results['demand_model']['verdict']}\n\n")
f.write("NOTE: Population figures are REAL (Dubai Statistics Center 2023). The\n")
f.write("participation rate and peaking factors are clearly-labeled planning\n")
f.write("assumptions, not measured data - stated so they can be challenged/tuned.\n")
print("Saved: catchment_demand_summary.txt")
```

Site Analysis & Human-Centric Research

What the site is, who it serves, and what is wrong with it

A flat, largely exposed 15,000 m² neighbourhood park serving roughly 7,640 residents within a ten-minute walk. Its defining problem is not layout or planting: for 55.5% of daylight hours, standing outside on it is uncomfortable or worse.

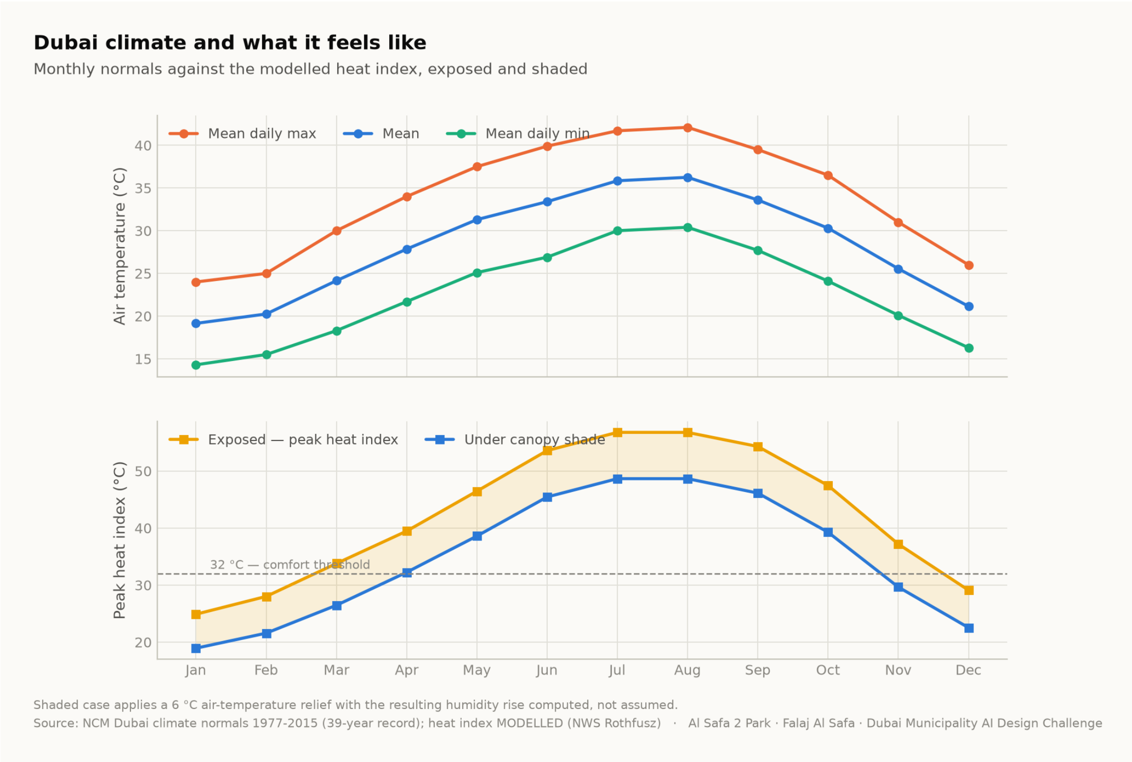
Upload slot 11 — Site Analysis & Human-Centric Research
Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai
Mohamed Wasim · Individual Applicant
Issued 01 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. Climate — the governing constraint

The analysis reconstructs an 8,760-hour year for the site from 39 years of National Centre of Meteorology monthly normals, with solar positions computed for every hour by the NREL algorithm at the site's coordinates. The reconstruction is verified back against the published normals to within 0.39 °C.

The climate series is modelled, not measured. The underlying record is 39 years long but is published as monthly normals; the hourly series is reconstructed from them and labelled as modelled everywhere it appears. Conclusions are about a typical year, never about extremes.



Dubai climate normals against the modelled heat index. Analysis output — computed from project data.

2. The problem, ranked rather than listed

Problem	Severity	Evidence
Thermal discomfort	Critical	Only 44.5% of 4,402 daylight hours are comfortable; peak heat index 56.8 °C
No continuous shaded route	Critical	A straight or fragmented shade strategy leaves 330 hours a year with no shade anywhere along the primary route
Exposed perimeter	High	Roads on the boundary contribute noise, glare and reflected heat with no intervening mass
Water and irrigation demand	High	Open water or irrigated turf in this climate carries a continuing evaporation and supply cost
Limited programme range	Moderate	Without comfort, no amount of programming produces use

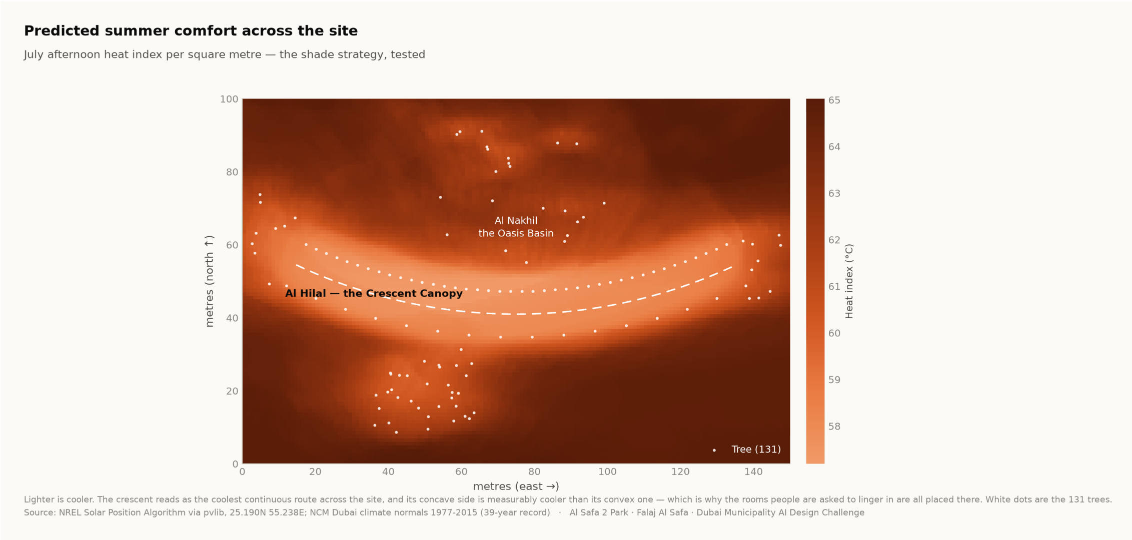
Ranking by severity is what justifies making shade the organising idea rather than one strategy among several.

3. Who the park serves

Approximately 7,640 residents live within a ten-minute walk (Dubai Statistics Center, 2023). The neighbourhood is established and low-rise, with families, working adults, domestic staff and a significant older population. The user groups and their needs are set out in the User Experience and Activation Strategy.

Visitor demand is a scenario, not a prediction. No footfall data exists for this site. Demand figures are deliberately excluded from the machine learning suite, because a model trained on an assumed demand curve would only recover the assumption that produced it.

4. Where the site is comfortable, square metre by square metre



Predicted July afternoon heat index per m². The crescent reads as the coolest continuous route, and its concave side is measurably cooler than its convex face. Analysis output — computed from project data.

5. Opportunities the analysis identifies

- **One continuous shaded route is worth more than dispersed shade.** Permutation importance puts distance to the crescent far above every other geometric feature.
- **Comfort is predictable from the calendar.** At 97.5% accuracy from sun position and date alone, park operations need no sensor network.
- **The shoulder seasons are recoverable.** The comfort gain concentrates in late afternoon in spring and autumn — hours that are currently lost and are cheap to win back.
- **The perimeter is an asset.** A planted berm addresses noise, glare and heat simultaneously.

Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.

Fig01 climate and comfort

Analysis output — computed from project data.

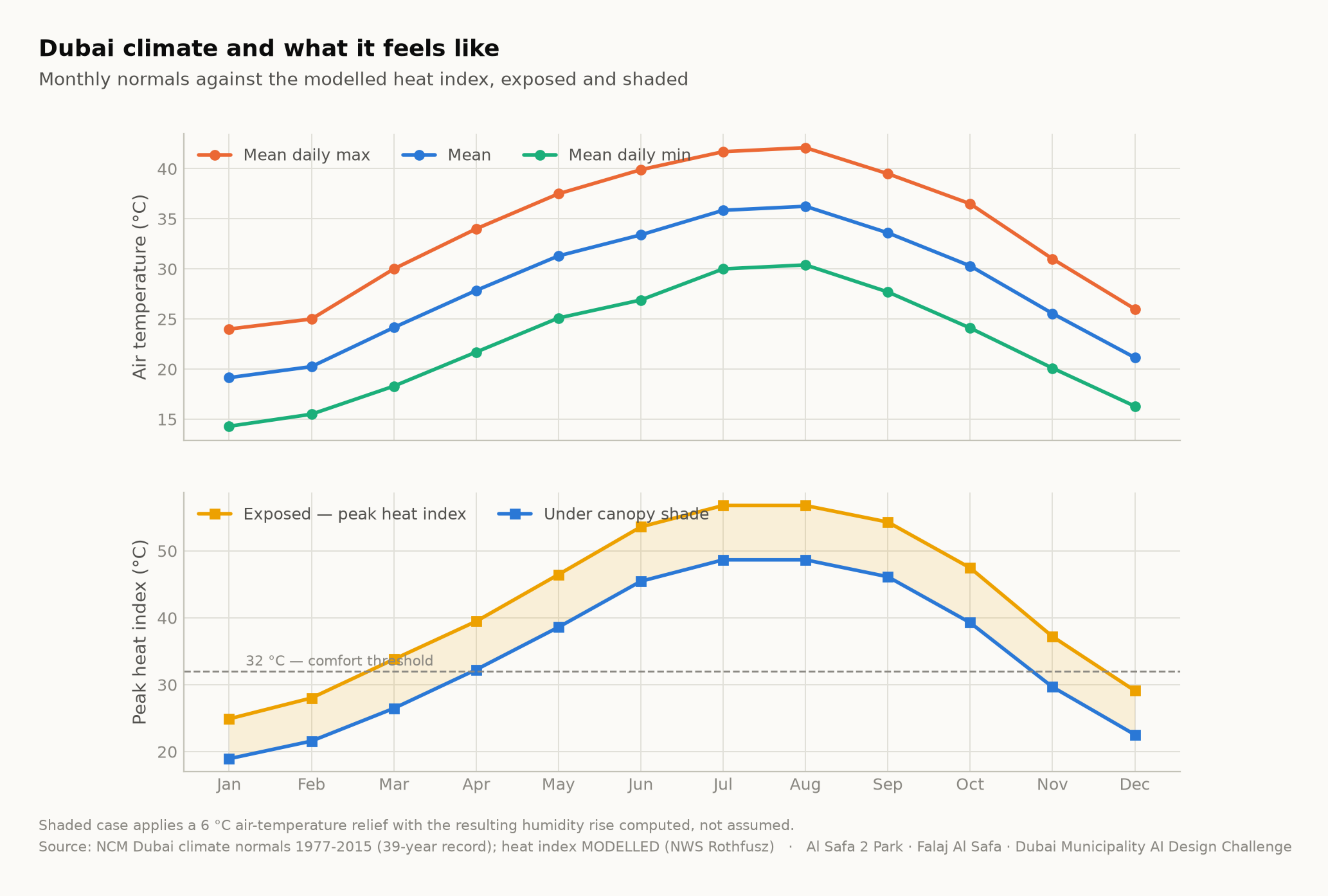
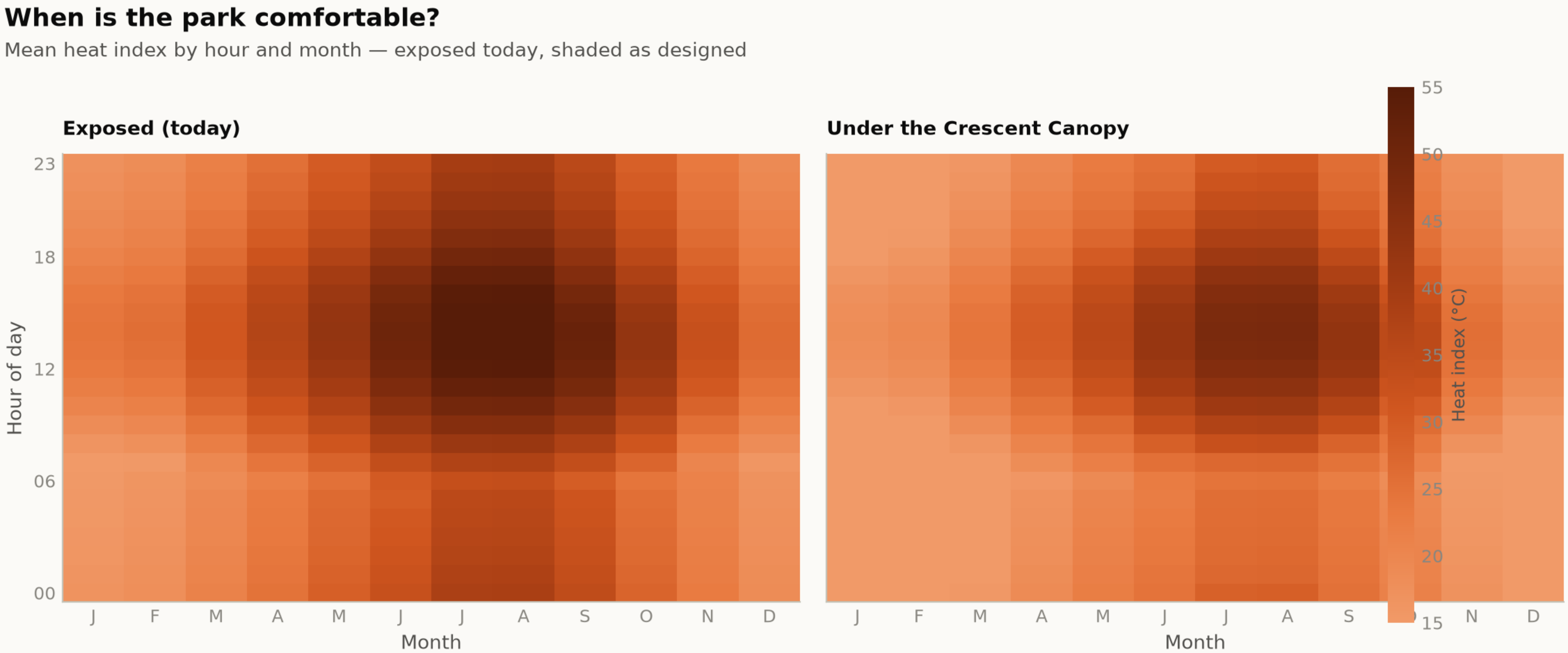


Fig09 diurnal comfort

Analysis output — computed from project data.



The design opens up the late afternoon in spring and autumn, which is when the catchment most wants to use the park.

Source: NCM Dubai climate normals 1977-2015 (39-year record) — hourly series MODELLED, see src/climate.py · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge